

3.13.2 Analogue and digital signals (A-level only)

3.13.2.1 Difference between analogue and digital signals (A-level only)

Content

Bits, bytes.

Analogue-to-digital conversion:

- sampling audio signals for transmission in digital form
- conversion of analogue signals into digital data using two voltage levels
- quantisation
- sampling rate
- effect of sampling rate and number of bits per sample on quality of conversion
- advantages and disadvantages of digital sampling
- process of recovery of original data from noisy signal
- effect of noise in communication systems.

Pulse code modulation.

Students should appreciate the use of a variety of sensors to collect analogue data.

The ability to carry out binary arithmetic is not required. Knowledge of binary numbers 1 to 10 is adequate.